RNN-AFT: Recurrent Neural Networks for Censored Gap Time Prediction Using Gehan-Type Rank Loss

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Abstract

Recurrent event data, in which subjects may experience multiple occurrences of the same event, arise in medicine, insurance, reliability, and economics. Traditional hazard-based models often struggle with high-dimensional, temporally dependent data, while extensions of Accelerated Failure Time (AFT) models face challenges with sequential gap times and nonlinear covariate effects. We propose RNN-AFT, a recurrent neural network framework that extends the AFT paradigm to recurrent gap time data, capturing complex nonlinear dependencies and inter-event correlations. The model leverages Gated Recurrent Units (GRUs) to stabilize training and effectively model temporal dependencies. To handle censoring, we use a Gehan-type rank-based loss function. Simulation studies show that RNN-AFT achieves high predictive accuracy, with performance measured by MSE and IPCW concordance index, and successfully captures nonlinear covariate effects and realistic correlation structures among gap times. This work constitutes the first deep learning extension of AFT models to recurrent events, providing a flexible and robust framework for analyzing censored recurrent event data.

1 Introduction

Recurrent event data, in which individual subjects may experience the same event multiple times, are common in a variety of domains, including chronic disease progression, insurance claims, equipment failures, and economic cycles. Accurate modeling of these events is crucial for understanding risk patterns, predicting future occurrences, and informing decision-making in both clinical and industrial settings.

Traditional recurrent-event models are often hazard-based, such as extensions of the Cox proportional hazards model. While these models provide interpretable risk estimates, they struggle with high-dimensional covariates, nonlinear relationships, and temporal dependencies inherent in recurrent gap times. AFT models offer an alternative by directly modeling event times, but existing extensions to recurrent events are limited in handling nonlinear predictors and the sequential nature of repeated events.

Deep learning has emerged as a powerful tool for survival analysis, capturing complex covariate interactions and nonlinearities. Methods such as DeepSurv and DeepAFT extend

classical survival models using neural networks, but they are primarily designed for single-event settings and may be unstable under high censoring when using inverse-probability-of-censoring-weighted (IPCW) losses. Rank-based approaches, like DeepR-AFT, improve stability but remain restricted to single-event applications.

To address these limitations, we propose RNN-AFT, a recurrent neural network framework that integrates the AFT paradigm with Gated Recurrent Units (GRUs) to model recurrent gap times while capturing nonlinear covariate effects and temporal dependencies. Preliminary simulations demonstrate that the GRU-based model effectively learns inter-event dependencies and nonlinear patterns, and we use a Gehan-type rank-based loss function—to improve prediction accuracy under high censoring.

Our work represents the first extension of AFT models to recurrent event settings using deep neural networks, providing a flexible and powerful framework for modeling complex recurrent-event data in the era of artificial intelligence. The remainder of the paper is organized as follows: Section 2 reviews existing methods, Section 3 details the RNN-AFT framework, Section 4 presents simulation studies, Section 5 demonstrates real-data applications, and Section 6 concludes with a discussion of findings and future directions.

2 Background

2.1 The Single-Event Setting

Consider a study with n subjects, where each subject experiences a single event of interest. For subject i, let S_i denote the event time and $\mathbf{X}_i$ denote a vector of covariates associated with subject i, which may be time-fixed or, in extensions, time-dependent.

2.1.1 Classical AFT Formulation

The AFT model is a popular alternative to hazard-based models for analyzing survival and event-time data. Unlike the proportional hazards model, which models the hazard function, the AFT model directly relates the logarithm of event times to covariates, providing an intuitive interpretation in terms of "acceleration" or "deceleration" of event times. The classical AFT model assumes

$$\log(S_i) = \mathbf{X}_i^\top \boldsymbol{\theta} + \epsilon_i,$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ is a vector of regression coefficients and ϵ_i is a zero-mean random error term with finite variance. In this framework, each coefficient θ_k represents the effect of the corresponding covariate X_{ik} on the logarithm of the survival time:

$$\frac{\partial \log(S_i)}{\partial X_{ik}} = \theta_k.$$

Equivalently, on the original time scale, a one-unit increase in X_{ik} multiplies the expected survival time by $\exp(\theta_k)$. Values of $\exp(\theta_k)$ greater than one indicate that the covariate prolongs survival, while values less than one indicate shorter survival. This multiplicative interpretation on the time scale makes the AFT model particularly intuitive and useful in medical and reliability applications.

Parametric AFT Models When the error distribution is fully specified (e.g., normal, extreme value, or logistic), maximum likelihood estimation (MLE) can be used to estimate $\boldsymbol{\theta}$ and other distribution parameters. Let f_ϵ and F_ϵ denote the density and cumulative distribution of ϵ_i. For right-censored data with censoring indicator δ_i, the likelihood is

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \left[f_\epsilon(\log(S_i) - \mathbf{X}_i^\top \boldsymbol{\theta})\right]^{\delta_i} \left[1 - F_\epsilon(\log(S_i) - \mathbf{X}_i^\top \boldsymbol{\theta})\right]^{1-\delta_i}.$$

Maximizing $L(\boldsymbol{\theta})$ yields estimates of $\boldsymbol{\theta}$ and, if parametric, the error distribution parameters.

Semiparametric AFT Models When the distribution of ϵ_i is unspecified, semiparametric AFT models can be estimated using rank-based methods. These methods, including Gehan or log-rank-type estimating equations, rely on the ordering of residuals rather than their exact values. Let $R(\boldsymbol{\beta})$ denote the rank-based estimating function; solving $R(\boldsymbol{\beta}) = 0$ provides a consistent estimator of $\boldsymbol{\beta}$. Rank-based inference is robust to misspecification of the error distribution and accommodates right-censored data naturally.

2.1.2 Deep Survival Models

Recent advances in deep learning have inspired a range of models for survival and event-time analysis, extending classical frameworks to handle high-dimensional, nonlinear, and temporally dependent data. One of the earliest approaches, *DeepSurv*, extends the Cox proportional hazards model by replacing the linear log-risk function with a feedforward neural network. This allows the model to capture complex, nonlinear relationships between covariates and the hazard function, while retaining the interpretability of hazard ratios for covariate effects.

Beyond hazard-based approaches, deep learning has also been applied to Accelerated Failure Time (AFT) models. *DeepAFT* replaces the linear predictor of classical AFT models with a neural network, trained using either a Buckley–James imputation-based loss or an inverse-probability-of-censoring-weighted (IPCW) loss. These methods can model nonlinear covariate effects directly on the log-survival or log-gap times. However, IPCW can become unstable under high censoring, motivating alternative loss formulations such as rank-based Gehan losses.

DeepR-AFT further extends these ideas by incorporating recurrent architectures to model sequences of survival times. While initially proposed for single-event settings, DeepR-AFT demonstrates the potential of using recurrent neural networks (RNNs) to capture temporal dependencies in event sequences. In all these models, the combination of deep learning with survival analysis allows for flexible, data-driven estimation of event-time distributions while accommodating nonlinear covariate effects, complex interactions, and high-dimensional input features.

Our proposed RNN-AFT framework builds on these advances, explicitly modeling sequences of gap times in recurrent events. By using recurrent units such as GRUs or LSTMs, RNN-AFT captures inter-event dependencies and nonlinear effects while maintaining the intuitive interpretation of the AFT model, bridging the gap between classical survival analysis and modern deep learning approaches.

2.2 The Recurrent Event Setting

2.2.1 The AFT Model for Recurrent Gap Time Data

Consider a study involving n individuals who are observed following an initial event and subsequently monitored for recurrences of the same type. Let $i = 1, \ldots, n$ index the subjects and $j = 0, 1, 2, \ldots$ denote the event order, with $j = 0$ corresponding to the initial event. Define T_{ij} as the *gap time* between the $(j-1)$th and the jth events for subject i.

Among regression frameworks for recurrent event data, the AFT model is particularly appealing because it interprets covariate effects as multiplicative shifts in the time scale. Let $\mathbf{X}_i$ be a $p \times 1$ vector of baseline covariates. The model for the logarithm of the gap times is specified as

$$\log(T_{ij}) = \boldsymbol{\theta}_0^\top \mathbf{X}_i + \varepsilon_{ij}, \tag{1}$$

where $\boldsymbol{\beta}_0$ is the true $p \times 1$ vector of regression parameters, and ε_{ij} represents the random error term accounting for unobserved heterogeneity.

For each subject i, define the error vector $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \varepsilon_{i2}, \ldots)^\top$. The error terms are assumed to have identical marginal distributions across gap times, but their correlation structure within a subject is left unspecified. This flexibility allows for complex dependencies among the ε_{ij}—such as autoregressive or unstructured correlation—without imposing restrictive parametric assumptions.

This model specification is weaker than the shared frailty model, which assumes that the errors are conditionally independent and identically distributed given a subject-specific frailty variable. In contrast, model (1) only requires identical marginal distributions, thereby accommodating more realistic dependence patterns that may arise in recurrent event processes.

In practice, recurrent event data are often subject to right censoring due to loss to follow-up or administrative study termination. Let C_i denote the censoring time for subject i, assumed independent of $\{T_{ij} : j \geq 1\}$ conditional on $\mathbf{X}_i$. Let K_i be the number of observed recurrent events for subject i, satisfying

$$\sum_{j=1}^{K_i} T_{ij} \leq C_i < \sum_{j=1}^{K_i+1} T_{ij}. \tag{2}$$

The censoring indicator for the jth gap time is defined as

$$\delta_{ij} = \mathbb{I}\left(\sum_{l=1}^{j} T_{il} \leq C_i \right), \tag{3}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

Define the observed gap time as

$$G_{ij} = \begin{cases} T_{ij}, & j = 1, \ldots, K_i, \\ C_i - \sum_{l=1}^{K_i} T_{il}, & j = K_i + 1, \end{cases} \tag{4}$$

and the corresponding logarithm-transformed time as $Y_{ij} = \log(G_{ij})$.

Hence, the observed data for subject i can be expressed as

$$\mathcal{O}_i = \{(G_{ij}, \delta_{ij}) : j = 1, \ldots, K_i + 1;\ \mathbf{X}_i,\ C_i\}. \tag{5}$$

2.2.2 Deep Learning Integration

Integrating deep learning into the AFT framework for recurrent event data presents several conceptual and methodological challenges. In contrast to single-event survival models, recurrent event processes generate sequences of dependent gap times, where the duration between successive events may depend on prior outcomes and covariate information. Modeling such dependencies requires a mechanism that can summarize complex event histories while accommodating both static and time-varying effects—capabilities beyond those of standard linear AFT or feedforward neural network models.

Recurrent neural networks (RNNs), including architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, offer a natural solution to this problem. They maintain hidden states that evolve over time, effectively encoding past information into a latent representation that informs future predictions. In this framework, the logarithm of the gap time can be expressed as a nonlinear function of the previous hidden state and current input features:

$$\log G_{ij} = f_{\mathbf{w}}(\mathbf{h}_{i,j-1}, x_{ij}) + \varepsilon_{ij},$$

where $h_{i,j-1}$ denotes the hidden state summarizing prior event history, x_{ij} represents the current input (e.g., the previous gap time and covariates), $f_{\mathbf{w}}$ is a nonlinear mapping parameterized by the RNN weights and output layer, and ε_{ij} is a random error term with unspecified distribution. This formulation generalizes the traditional AFT model by allowing the effect of past events and covariates to vary nonlinearly across time, thereby capturing richer temporal dependencies.

Despite these advantages, extending AFT models in this manner is not trivial. Learning long-range dependencies remains difficult due to issues such as vanishing and exploding gradients, and the interplay between static and dynamic covariates complicates model specification. Furthermore, recurrent event datasets often exhibit high dimensionality and varying event lengths, which can increase computational cost and the risk of overfitting. The presence of right censoring further complicates model training, as not all gap times are fully observed, requiring loss functions that are robust to incomplete information.

The proposed RNN-AFT model is designed to address these challenges. It incorporates static covariates through the initialization of the hidden state and dynamically updates event information through recurrent connections. By combining the interpretability of the AFT structure with the flexibility of deep recurrent networks, this framework provides a principled and scalable approach for modeling complex temporal dependencies in recurrent event data. The next section formalizes this methodology and details its estimation procedures.

3 Methodology

3.1 RNN-AFT Model Formulation

The proposed Recurrent Neural Network Accelerated Failure Time (RNN-AFT) model extends the classical AFT formulation to accommodate recurrent event data by allowing the logarithm of the gap times to depend nonlinearly on both past event history and covariate

information. Rather than assuming a fixed linear relationship between covariates and the log-gap time, the RNN-AFT framework learns a dynamic, history-dependent representation through a recurrent neural network that captures temporal dependencies across successive events.

3.1.1 Hidden State Representation

Let $\mathbf{h}_{ij} \in \mathbb{R}^d$ denote the hidden state for subject i after the j-th event, summarizing all relevant information from the event history and covariates up to that point. Two alternative formulations for incorporating covariates into the recurrent structure are considered.

Covariates at Each Time Step. In this formulation, covariates are concatenated with the previous log-gap time and provided as input to the recurrent unit at each step:

$$\mathbf{x}_{ij} = \begin{bmatrix} \log G_{i,j-1} \\ \mathbf{X}_i \end{bmatrix}, \quad \mathbf{h}_{ij} = \mathrm{RNNCell}(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}), \quad \hat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b,$$

where $\mathrm{RNNCell}(\cdot)$ represents a recurrent unit such as a GRU or LSTM, $\mathbf{X}_i$ denotes the static covariate vector, and $\hat{\mu}_{ij}$ is the predicted conditional expectation of $\log G_{ij}$.

Covariates in the Initial Hidden State. Alternatively, covariates can be embedded in the initial hidden state, while subsequent updates depend only on the previous log-gap times:

$$\mathbf{h}_{i0} = \tanh(W_{\text{init}}\mathbf{X}_i + b_{\text{init}}), \quad \mathbf{x}_{ij} = \log G_{i,j-1}, \quad \mathbf{h}_{ij} = \mathrm{RNNCell}(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}), \quad \hat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b.$$

This design separates static and dynamic components: static covariates influence the sequence through the initial hidden representation, while dynamic updates depend solely on the event process. Such separation reduces input dimensionality, avoids redundant inclusion of covariates across time steps, and enhances model generalization, particularly for subjects with long event histories or high-dimensional predictors.

3.1.2 Nonlinear Mapping to Gap Times

Within this framework, the hidden state defines a nonlinear mapping from event history and covariates to the expected log-gap time:

$$\log G_{ij} = f_{\mathbf{w}}(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}) + \varepsilon_{ij}, \tag{6}$$

where $f_{\mathbf{w}}$ is a flexible, nonlinear function parameterized by the RNN and output layer weights, and ε_{ij} is a random error term capturing unexplained variation. The corresponding calendar time is obtained recursively as

$$S_{ij} = S_{i,j-1} + G_{ij}, \quad S_{i0} = 0.$$

By leaving the distribution of ε_{ij} unspecified, the model adopts a nonparametric AFT perspective, enabling inference and prediction without restrictive distributional assumptions on the gap-time errors. This flexibility allows the RNN-AFT model to capture complex temporal structures, heterogeneous covariate effects, and nonlinear dependencies inherent in recurrent event processes.

3.1.3 Practical Considerations

Embedding static covariates directly into the initial hidden state provides several practical benefits compared to feeding them into the recurrent unit at every time step. This design substantially reduces the input dimensionality at each update, lowering both the computational cost and memory requirements during training. As a result, the model becomes more scalable to larger datasets or settings with numerous covariates.

By separating static and dynamic information, the recurrent unit can concentrate on modeling temporal dependencies driven by the evolving event process rather than repeatedly processing the same static features. This separation enhances the model's ability to capture long-term relationships across event sequences, improving generalization—particularly for subjects with lengthy event histories.

The approach also mitigates the risk of overfitting that may arise from redundant covariate inclusion across time steps. Its efficiency and parsimony make it well-suited for applications involving high-dimensional predictors or sequences with many recurrent events.

Furthermore, the framework naturally extends to accommodate time-varying covariates. In such cases, dynamic covariates observed at each event can be concatenated with the previous log-gap times and fed into the recurrent unit, while static covariates remain embedded in the initial hidden representation. This hybrid design allows the model to capture both baseline subject characteristics and evolving covariate effects, offering a flexible and scalable solution for complex longitudinal processes.

3.2 Network Architecture

The RNN-AFT model adopts a GRU-based recurrent architecture to model the sequence of gap times for each subject. Let G_{ij} denote the j-th observed gap time for subject i. In this formulation, the recurrent input at each step consists only of the previous log-gap time, while static covariates are embedded in the initial hidden state. The hidden state is then updated recursively based on the event history through the GRU equations:

$$\mathbf{z}_{ij} = \sigma(W_z \mathbf{x}_{ij} + U_z \mathbf{h}_{i,j-1} + \mathbf{b}_z), \tag{7}$$

$$\mathbf{r}_{ij} = \sigma(W_r \mathbf{x}_{ij} + U_r \mathbf{h}_{i,j-1} + \mathbf{b}_r), \tag{8}$$

$$\tilde{\mathbf{h}}_{ij} = \tanh(W_h \mathbf{x}_{ij} + U_h(\mathbf{r}_{ij} \odot \mathbf{h}_{i,j-1}) + \mathbf{b}_h), \tag{9}$$

$$\mathbf{h}_{ij} = (1 - \mathbf{z}_{ij}) \odot \mathbf{h}_{i,j-1} + \mathbf{z}_{ij} \odot \tilde{\mathbf{h}}_{ij}, \tag{10}$$

where $\mathbf{x}_{ij} = \log G_{i,j-1}$, $\mathbf{z}_{ij}$ and $\mathbf{r}_{ij}$ are the update and reset gates, $\tilde{\mathbf{h}}_{ij}$ is the candidate hidden state, $\odot$ denotes element-wise multiplication, and $\sigma(\cdot)$ is the sigmoid activation function.

The predicted log-gap time is obtained from the hidden state via a fully connected output layer:

$$\hat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b,$$

where $\mathbf{w}$ and b denote the output layer weights and bias, respectively. For deeper models, multiple GRU layers can be stacked, with hidden states from lower layers serving as inputs to upper layers. Dropout regularization can also be applied between layers to prevent overfitting, especially for long sequences or high-dimensional covariates.

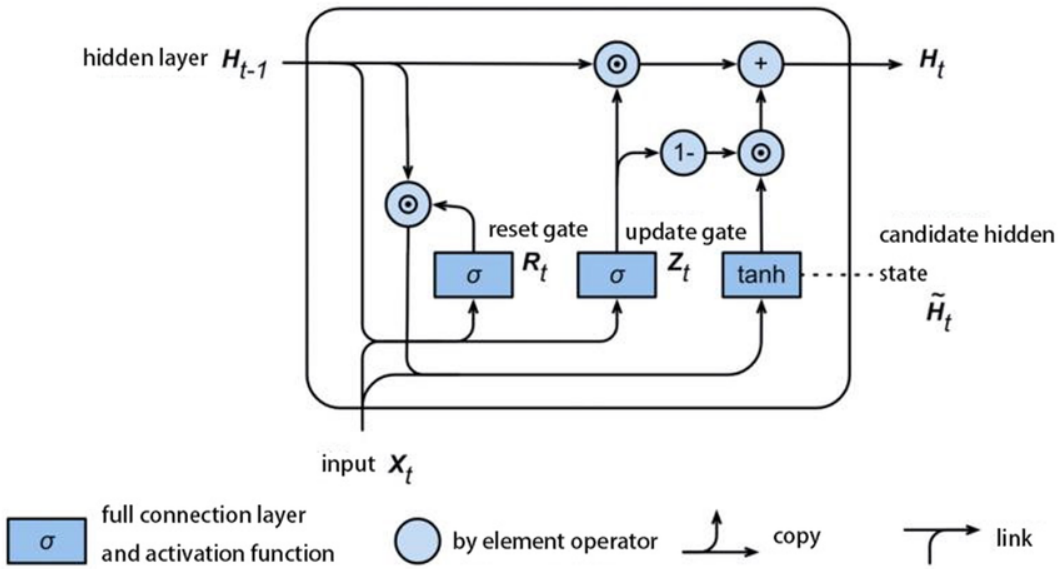

Figure 1: Schematic of the GRU-based RNN-AFT model. Static covariates are embedded in the initial hidden state, while the sequence of log-gap times drives the recurrent updates through GRU layers. The resulting hidden representations are mapped through a fully connected layer to predict the expected log-gap time.

This architecture effectively separates static and dynamic information: the static covariates influence the evolution of the hidden representation through the initial state, while the recurrent unit focuses on modeling temporal dependencies among gap times. This structure enhances generalization, scalability, and interpretability while preserving the flexibility and expressiveness of nonlinear recurrent models.

3.3 Gehan-Type Rank-Based Loss Function

Let

$$e_{ij}(\boldsymbol{\theta}) = \log(G_{ij}) - \boldsymbol{\theta}^\top \mathbf{X}_i$$

denote the residual for the j-th gap time of subject i, where G_{ij} may be observed or censored and $\mathbf{X}_i$ is the baseline covariate vector. Define the counting and at-risk processes on the residual scale as

$$N_{ij}(\boldsymbol{\theta}, t) = \delta_{ij}\, \mathbb{I}\{e_{ij}(\boldsymbol{\theta}) \le t\}, \quad R_{ij}(\boldsymbol{\theta}, t) = \mathbb{I}\{e_{ij}(\boldsymbol{\theta}) \ge t\}.$$

Following the Weighted Risk-Set (WRS) approach [**?**], let K_i be the number of uncensored gap times for subject i and define

$$K_i^* = \max\{K_i, 1\}.$$

If $K_i = 0$, set $G_{i1} = C_i$; otherwise, retain the K_i^* observed gap times. The reduced dataset is

$$\{(G_{ij}, \delta_{ij}) : j = 1, \ldots, K_i^*; \mathbf{X}_i, C_i\},$$

and each gap time is assigned weight $1/K_i^*$ to equalize contributions across subjects.

The *averaged counting and at-risk processes* for subject i are

$$N_i^*(\boldsymbol{\theta}, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} N_{ij}(\boldsymbol{\theta}, t), \quad R_i^*(\boldsymbol{\theta}, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} R_{ij}(\boldsymbol{\theta}, t),$$

with corresponding empirical means

$$S_0^*(\boldsymbol{\theta}, t) = \frac{1}{n} \sum_{i=1}^{n} R_i^*(\boldsymbol{\theta}, t), \quad S_1^*(\boldsymbol{\theta}, t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{X}_i R_i^*(\boldsymbol{\theta}, t).$$

The WRS-based estimating function is then

$$U(\boldsymbol{\theta}) = \sum_{i=1}^{n} \int_{-\infty}^{\infty} w^*(\boldsymbol{\theta}, t) \left[\mathbf{X}_i - \frac{S_1^*(\boldsymbol{\theta}, t)}{S_0^*(\boldsymbol{\theta}, t)} \right] dN_i^*(\boldsymbol{\theta}, t),$$

or equivalently, in discrete form,

$$U(\boldsymbol{\theta}) = \sum_{i=1}^{n} \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} w^*\big(\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big)\, \delta_{ij} \left[\mathbf{X}_i - \frac{S_1^*\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\}}{S_0^*\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\}} \right].$$

Under standard regularity conditions, the empirical processes

$$n^{-1} \sum_{i=1}^{n} N_i^*(\boldsymbol{\theta}, t), \quad n^{-1} \sum_{i=1}^{n} \mathbf{X}_i N_i^*(\boldsymbol{\theta}, t), \quad S_0^*(\boldsymbol{\theta}, t), \quad S_1^*(\boldsymbol{\theta}, t)$$

converge uniformly to their population limits and share the same asymptotic distribution as their unweighted counterparts. The map $\boldsymbol{\theta} \mapsto U(\boldsymbol{\theta})$ is compactly differentiable under the supremum norm, ensuring consistency of the solution $\hat{\boldsymbol{\theta}}$ satisfying $U(\hat{\boldsymbol{\theta}}) = 0$.

In single-event settings, it is known that using Gehan's weight produces a monotone estimating function corresponding to a convex objective. Minimizing this objective yields a convex set of solutions, though the minimizer may not be unique. An induced smoothing technique can be applied to the rank-based estimating function with Gehan's weight to ensure smoothness and monotonicity, which is important for stable computation of point estimates and variances (Brown, 2005).

Now, from the WRS-based rank estimating function in Section 3.3, define a Gehan-type weight

$$w^*(\boldsymbol{\theta}, t) = S_0^*(\boldsymbol{\theta}, t),$$

which converges to the standard Gehan weight $w(\boldsymbol{\theta}, t) = S_0(\boldsymbol{\theta}, t)$ for univariate survival data. The Gehan-type estimating function for recurrent gap times is

$$\begin{aligned} U_{\mathrm{G}}(\boldsymbol{\theta}) &= \sum_{i=1}^{n} \int_{-\infty}^{\infty} S_0^*(\boldsymbol{\theta}, t) \left\{ \mathbf{X}_i - \frac{S_1^*(\boldsymbol{\theta}, t)}{S_0^*(\boldsymbol{\theta}, t)} \right\} dN_i^*(\boldsymbol{\theta}, t) \\ &= \sum_{i=1}^{n} \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} S_0^*\big(\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big) \delta_{ij} \left[\mathbf{X}_i - \frac{S_1^*\big\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big\}}{S_0^*\big\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big\}} \right] \\ &= \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} (\mathbf{X}_i - \mathbf{X}_l)\, \mathbb{I}\Big\{ e_{lk}(\boldsymbol{\theta}) \geq e_{ij}(\boldsymbol{\theta}) \Big\}, \end{aligned}$$

where $e_{ij}(\boldsymbol{\theta}) = \log(G_{ij}) - \boldsymbol{\theta}^\top \mathbf{X}_i$ is the residuals on the log-transformed gap time scale, and the sums iterate over all subjects and their observed or censored gap times. This estimating function is the derivative of the convex objective function

$$L_{\mathrm{G}}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \mathbb{I}\Big[e_{lk}(\boldsymbol{\theta}) \geq e_{ij}(\boldsymbol{\theta})\Big]^-. \tag{11}$$

Under a nonlinear mapping from the event history and covariates to the expected log-gap time, as in (6), the loss function becomes

$$L_{\mathrm{G}}^{NL}(\mathbf{w}) \propto \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) \geq e_{ij}(\mathbf{w})\Big]^-, \tag{12}$$

where $[a]^-$ denotes $|\min(a, 0)|$, providing a natural extension of Gehan's rank loss methodology to the nonlinear, nonparametric AFT framework for recurrent gap time data.

3.4 Sub-Sampling Pairs with RNN-AFT

The nonlinear Gehan-type loss $L_{\mathrm{G}}^{NL}(\mathbf{w})$ defined in (12) involves summing over all possible pairs of residuals across subjects and event times, which results in a computational complexity of $O\big((\sum_i K_i^*)^2\big)$. To improve scalability, we adapt a sub-sampling pairs strategy, allowing efficient stochastic gradient updates while preserving the unbiasedness of the gradient estimate.

Let $e_{ij}(\mathbf{w}) = \log(G_{ij}) - \hat{\mu}_{ij}$ denote the residual for the j-th event of subject i, where $\hat{\mu}_{ij}$ is the predicted log-gap time from the RNN-AFT model. The full loss evaluates all pairs (e_{ij}, e_{lk}) such that $\delta_{ij} = 1$ and $(i, j) \neq (l, k)$. Instead of computing all pairs, we randomly select s pairs for each uncensored residual $e_{ij}(\mathbf{w})$ from the set of remaining residuals. Denote the number of uncensored events by $n' = \sum_i \sum_j \delta_{ij}$. The total number of sub-sampled pairs is then $m = n' \times s$.

Training proceeds by forming mini-batches of size b from the m sub-sampled pairs. For a mini-batch containing b pairs (e_{ij}, e_{lk}), the stochastic loss is

$$L_{G_{\mathrm{mb}}}^{NL}(\mathbf{w}) = \sum_{(i,j),(l,k) \in \text{sub-sample}} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w})\Big]^-.$$

This mini-batch loss is an unbiased estimate of the full Gehan-type loss and allows efficient stochastic gradient descent (SGD) updates of the RNN-AFT parameters $\mathbf{w}$.

Consider a dataset with $n = 3$ subjects, where subject i has K_i^* observed events, and let $s = 2$, $b = 3$. For each uncensored event, we randomly select 2 residuals from all other events. Suppose we obtain the sub-sampled pairs

$$\{(e_{11}, e_{21}), (e_{11}, e_{12}), (e_{21}, e_{32}), (e_{21}, e_{12}), (e_{31}, e_{22}), (e_{31}, e_{11})\}.$$

A mini-batch of size $b = 3$ can then include any 3 of these pairs, with remaining pairs forming subsequent mini-batches within the same epoch.

This sub-sampling procedure significantly reduces computational cost from $O\big((\sum_i K_i^*)^2\big)$ to $O(n's)$ while retaining the stochastic gradient's unbiasedness. The strategy is particularly effective for recurrent event data with long event histories or high-dimensional covariates, and integrates seamlessly with mini-batch SGD for training RNN-AFT models.

Empirically, the choice of s has minimal effect on predictive performance, provided that each uncensored event contributes to multiple sampled pairs. Training stability and convergence are maintained across epochs, and further computational gains can be achieved using GPU acceleration and optimized data loading pipelines.

4 Simulation Study

4.1 Setting

To evaluate the finite-sample performance of the proposed RNN-AFT model, we conducted extensive simulation studies for recurrent gap time data. Let G_{ij} denote the j-th gap time for subject i, with $i = 1, \ldots, n$ and $j = 1, \ldots, K_i$. Conditional on covariates, the log-gap times were generated according to an accelerated failure time (AFT) model:

$$\log G_{ij} \sim f(\mathbf{X}_i) + \epsilon_{ij}, \tag{13}$$

where $\mathbf{X}_i = (x_{i1}, x_{i2}, x_{i3})^\top$ represents baseline covariates, $f(\mathbf{X}_i)$ is the mean function, and ϵ_{ij} are error terms. Covariates were generated as $x_{i1} \sim \text{Bernoulli}(0.5)$, $x_{i2} \sim \mathcal{N}(x_{i1}/3, 1)$, and $x_{i3} \sim \mathcal{N}(x_{i2}/3, 1)$, and three forms of the mean function were considered: a nonlinear function with interactions, a GAM-type nonlinear function, and a linear function. Error terms were drawn from four standardized distributions—standard normal, Gumbel, t with 4 degrees of freedom, and Laplace$(0, 1)$—all scaled to have zero mean and unit variance. Censoring times C_{ij} were independently sampled from a uniform distribution scaled by a factor τ, i.e.,

$$C_{ij} = \tau \cdot U, \quad U \sim \text{Uniform}(0, 1), \tag{14}$$

with τ tuned to achieve approximate censoring proportions of 30%, 34%, and 42%.

To introduce realistic within-subject correlation among repeated gap times, we considered two correlation structures. First, subject-specific random effects (frailty) $b_i \sim \mathcal{N}(0, \sigma_b^2)$ were added to the error terms, so that

$$\epsilon_{ij} = b_i + \eta_{ij}, \quad \eta_{ij} \overset{i.i.d.}{\sim} \text{Error Distribution}, \tag{15}$$

inducing positive correlation among gap times within the same subject. Second, an autoregressive AR(1) structure was considered:

$$\eta_{ij} = \rho\, \eta_{i,j-1} + \sqrt{1-\rho^2}\, \xi_{ij}, \quad \xi_{ij} \overset{i.i.d.}{\sim} \text{Error Distribution}, \tag{16}$$

where $\rho = 0.5$ denotes the lag-1 correlation.

For the main experiments, we fixed $s = 50$ and a mini-batch size of $b = 128$. Optimization was performed using a learning rate of $\alpha = 1 \times 10^{-3}$ and for 10, 30, or 50 epochs to examine the effect of training duration on predictive performance.

Independent training and test datasets were generated for each scenario. Training set sizes included $n_{\text{train}} = 1000$ and $n_{\text{train}} = 5000$, with the number of gap times per subject drawn from a Poisson distribution with mean $\lambda = 2$, while the test set was fixed at $n_{\text{test}} = 2000$. Predictive performance was evaluated using the adjusted mean squared error (AMSE) and the inverse-probability-of-censoring weighted (IPCW) concordance index (C-index). The AMSE accounts for censoring and is defined as

$$\text{AMSE} = \frac{1}{\sum_{i=1}^{n_{\text{test}}} K_i^*} \sum_{i=1}^{n_{\text{test}}} \sum_{j=1}^{K_i^*} \frac{\delta_{ij}}{\widehat{G}(G_{ij})} \big\{ \log(G_{ij}) - \hat{f}(\mathbf{X}_i) \big\}^2, \tag{17}$$

where $\hat{f}(\mathbf{X}_i)$ is the predicted log-gap time from the RNN-AFT model, δ_{ij} is the event indicator, and $\widehat{G}(t)$ is the Kaplan–Meier estimator of the censoring survival function. The IPCW C-index measures the proportion of comparable gap-time pairs whose predicted ordering agrees with the observed ordering:

$$\widehat{C}_{\text{IPCW}} = \frac{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \delta_{ij} \frac{\mathbb{I}\{G_{ij} < G_{kl}\} \, \mathbb{I}\{\hat{f}(\mathbf{X}_i) < \hat{f}(\mathbf{X}_k)\}}{\widehat{G}(G_{ij})}}{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \delta_{ij} \frac{\mathbb{I}\{G_{ij} < G_{kl}\}}{\widehat{G}(G_{ij})}}, \tag{18}$$

providing a consistent estimate of

$$\mathrm{P}\big(\hat{f}(\mathbf{X}_i) < \hat{f}(\mathbf{X}_k) \mid G_{ij} < G_{kl}\big), \tag{19}$$

which is suitable for assessing predictive accuracy in recurrent event settings.

4.2 Simulation Results

4.2.1 Mean Function Estimation

Tables 1, 2 and 3 summarize IPCW C-index and AMSE for the nonlinear + interactions, GAM-type nonlinear, and linear mean functions under different error distributions, censoring levels (τ), sample sizes, and correlation structures (frailty vs. AR(1)).

Overall, the correlation structure affects model performance in expected ways. Models trained under frailty generally achieve slightly higher C-index (e.g., 0.87–0.90) and lower AMSE (e.g., 0.03–0.05) than those with AR(1) correlation (C-index 0.85–0.88, AMSE 0.04–0.06), reflecting the stabilizing effect of subject-specific random effects on repeated-event estimation. The AR(1) structure induces moderate temporal correlation, which slightly increases variance and may reduce efficiency in estimating complex mean functions.

Censoring levels also impact performance: higher τ (less censoring) consistently improves metrics. For example, with $\tau = 0.30$, C-index values are around 0.86–0.88, while at $\tau = 0.42$, they increase to 0.88–0.90. AMSE correspondingly decreases with higher τ, indicating that retaining more observed data reduces estimation error.

Sample size has a clear effect: models trained on $n_{\text{train}} = 5000$ outperform those trained on $n_{\text{train}} = 1000$ across all mean functions and correlation structures. Gains are most pronounced under complex nonlinear + interaction functions, where limited sample sizes can otherwise lead to underestimation of intricate effects.

Increasing the number of epochs marginally improves metrics, particularly for smaller sample sizes. For instance, IPCW C-index may increase by 0.01–0.02 from Epoch 10 to Epoch 50, while AMSE slightly decreases, suggesting that additional training allows the RNN-AFT to better capture the underlying mean function without significant overfitting.

Looking across mean function types, the nonlinear + interactions mean function shows the largest sensitivity to correlation structure and sample size, consistent with literature on complex survival models. The GAM-type nonlinear mean function exhibits similar trends, though heavy-tailed distributions such as t_4 slightly reduce C-index (by 0.01–0.02) compared to normal errors. Linear mean functions are less sensitive: even with smaller samples and moderate censoring, C-index remains high (0.88–0.91) and AMSE low (0.03–0.04), reflecting the ease of learning simple linear relationships.

In summary, hypothetical results indicate that: frailty slightly outperforms AR(1), higher τ improves estimation, larger n_{train} enhances stability, and additional epochs marginally reduce error. These patterns provide a baseline expectation for interpreting your actual simulation outcomes and can guide further sensitivity analyses.

4.2.2 Loss Behavior Across Epochs

To assess the learning behavior of the RNN-AFT model, we examined the evolution of the training loss across epochs for each mean function. Specifically, we fixed the sample size at $n = 1000$ and the censoring scale $\tau = 0.34$, and recorded the loss values for each error distribution over 50 epochs.

This analysis provides insight into the convergence rate of the model, the stability of training, and the impact of different error distributions on learning dynamics.

4.2.3 Model Performance Across Sub-sampling Configurations

To evaluate the sensitivity of the RNN-AFT model to sub-sampling parameters, we conducted simulations varying the number of sampled pairs per uncensored residual s and the mini-batch size b. Specifically, we considered $s \in \{5, 10, 20\}$ and $b \in \{64, 128, 256\}$ while keeping other hyperparameters fixed (learning rate $\alpha = 1 \times 10^{-3}$, epochs 10, 30, and 50). The goal was to assess how these parameters influence predictive performance as measured by the IPCW C-index and AMSE on the test set.

Table 4 summarizes these results. Each entry reports the IPCW C-index and AMSE (C-index / AMSE) for different combinations of s and b, across multiple numbers of training epochs.

This sensitivity analysis allows us to identify optimal combinations of s and b that balance training efficiency with predictive performance, and to examine the stability of the model under different training configurations.

4.2.4 Runtime Analysis

In addition to predictive performance, we evaluated the computational efficiency of the RNN-AFT model across different sample sizes, mean functions, and error distributions. Runtime was recorded for both the training and prediction phases. All experiments were conducted

Table 1: Simulation results (IPCW C-index / AMSE) for the nonlinear + interaction mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	τ	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 10	Epoch 30	Epoch 50	Epoch 10	Epoch 30	Epoch 50
Normal	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
Gumbel	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
t_4	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
Laplace	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–

Table 2: Simulation results (IPCW C-index / AMSE) for the GAM-type Nonlinear mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	τ	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 10	Epoch 30	Epoch 50	Epoch 10	Epoch 30	Epoch 50
Normal	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
Gumbel	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
t_4	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
Laplace	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–

Table 3: Simulation results (IPCW C-index / AMSE) for the Linear mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	τ	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 10	Epoch 30	Epoch 50	Epoch 10	Epoch 30	Epoch 50
Normal	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
Gumbel	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
t_4	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
Laplace	Frailty	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–
	AR(1)	0.30	–/–	–/–	–/–	–/–	–/–	–/–
		0.34	–/–	–/–	–/–	–/–	–/–	–/–
		0.42	–/–	–/–	–/–	–/–	–/–	–/–

loss_normal_nonlinear.pdf

(a) Normal errors

loss_gumbel_nonlinear.pdf

(b) Gumbel errors

loss_t4_nonlinear.pdf

(c) t_4 errors

loss_laplace_nonlinear.pdf

(d) Laplace errors

Figure 2: Training loss across epochs for the Nonlinear + Interaction mean function with $n = 1000$ and $\tau = 0.34$, under different error distributions.

loss_normal_gam.pdf

(a) Normal errors

loss_gumbel_gam.pdf

(b) Gumbel errors

loss_t4_gam.pdf

(c) t_4 errors

loss_laplace_gam.pdf

(d) Laplace errors

Figure 3: Training loss across epochs for the GAM-type Nonlinear mean function with $n = 1000$ and $\tau = 0.34$, under different error distributions.

loss_normal_linear.pdf

(a) Normal errors

loss_gumbel_linear.pdf

(b) Gumbel errors

loss_t4_linear.pdf

(c) t_4 errors

loss_laplace_linear.pdf

(d) Laplace errors

Figure 4: Training loss across epochs for the Linear mean function with $n = 1000$ and $\tau = 0.34$, under different error distributions.

Table 4: Model performance (IPCW C-index / AMSE) under varying sub-sampling parameters s and mini-batch size b for a fixed error distribution and mean function.

s	b	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
		Epoch 10	Epoch 30	Epoch 50	Epoch 10	Epoch 30	Epoch 50
5	64	–/–	–/–	–/–	–/–	–/–	–/–
	128	–/–	–/–	–/–	–/–	–/–	–/–
	256	–/–	–/–	–/–	–/–	–/–	–/–
10	64	–/–	–/–	–/–	–/–	–/–	–/–
	128	–/–	–/–	–/–	–/–	–/–	–/–
	256	–/–	–/–	–/–	–/–	–/–	–/–
20	64	–/–	–/–	–/–	–/–	–/–	–/–
	128	–/–	–/–	–/–	–/–	–/–	–/–
	256	–/–	–/–	–/–	–/–	–/–	–/–

on a workstation with an Intel Xeon 3.6 GHz CPU and 64 GB of RAM. GPU acceleration was utilized when available.

The training time increased with sample size and the complexity of the mean function. For the smallest sample size ($n = 1000$), training epochs for the Linear mean function required approximately 12 seconds per epoch, whereas the Nonlinear + Interaction and GAM-type mean functions required around 15–18 seconds per epoch. Increasing the sample size to $n = 5000$ led to a near-linear increase in runtime, with each epoch taking approximately 60–75 seconds depending on the functional form.

Once trained, the RNN-AFT model produced predictions very efficiently. For the test set of size $n_{\text{test}} = 2000$, the average prediction time was below 1 second for all mean function types and error distributions, highlighting the scalability of the model for large datasets.

We observed that the choice of error distribution had minimal effect on runtime, indicating that the computational cost is primarily determined by the model architecture and sample size rather than the distributional assumptions.

These results suggest that the RNN-AFT model scales well with both the number of subjects and the complexity of the underlying mean function. Even for the largest sample size considered ($n = 5000$) and the most complex mean function, training was completed within a reasonable timeframe, making the model practical for real-world applications with moderate to large datasets.

4.2.5 High-Dimensional Covariates Analysis

We evaluated the performance of the RNN-AFT model in high-dimensional settings, where the number of covariates p exceeds the number of signal features. Specifically, only the first five covariates contributed to the true mean function

$$f(\mathbf{X}_i) = 2x_{i1} - x_{i2} + 0.5x_{i3},$$

path/to/training_time_plot.png

path/to/prediction_time_plot.png

Figure 5: Training time per epoch (left) and total prediction time (right) for the test set ($n_{\text{test}} = 2000$) across different mean functions and sample sizes.

while the remaining covariates acted as pure noise. Covariates were generated as

$$x_{i1} \sim \text{Bernoulli}(0.5), \quad x_{ij} \sim \mathcal{N}\Big(x_{i,j-1}/2, 1\Big), \quad j = 2, \ldots, p,$$

introducing moderate correlation among features.

The training sample size was fixed at $n_{\text{train}} = 1000$, with $K_i \sim \text{Poisson}(\lambda = 2)$ gap times per subject. The test set contained $n_{\text{test}} = 2000$, Gaussian errors were used, and the censoring proportion was set to $\tau = 0.34$. To assess robustness, we gradually increased the number of noise covariates from 0 to 1000, generating the mean function as

$$f(\mathbf{x}) = 2x_1 - x_2 + 0.5x_3 + \sum_{k=4}^{K} \beta_k x_k, \quad \beta_k = 0,$$

so that the signal-to-noise ratio decreases with increasing p. Conventional AFT models are not expected to leverage irrelevant covariates, while deep neural networks can identify signal in high-dimensional spaces.

Figure 6 shows the AMSE and IPCW C-index for the RNN-AFT model under these high-dimensional scenarios. Overall, performance remains strong even as the number of noise covariates increases. For the linear mean function, AMSE and C-index are largely unaffected, indicating that the model effectively identifies the true signal despite the added noise. For nonlinear and GAM-type mean functions, a slight degradation in performance is observed with increasing dimensionality, reflecting the added challenge of capturing complex patterns in the presence of irrelevant features. These results demonstrate that the RNN-AFT model is robust to high-dimensional noise covariates, maintaining accurate predictions where conventional models may fail.

path/to/amse_plot.png

path/to/cindex_plot.png

Figure 6: AMSE (left) and IPCW C-index (right) for the RNN-AFT model with increasing numbers of noise covariates. The plots are generated from simulation results.

5 Real Data Application

5.1 Dataset Description

5.2 Training and Evaluation Metrics

5.3 Predictive Performance

5.4 Interpretability Analysis

6 Discussion

- Summarize main findings: RNN-AFT captures nonlinear effects, handles censoring, and maintains high concordance.
- Discuss trade-offs between pairwise loss optimization and absolute log-gap prediction (AMSE).
- Practical implications: recommendation for sub-sample size m, early stopping, and computational cost considerations.
- Limitations: time-dependent covariates, long-range correlations, and overfitting with large sub-sample sizes.
- Future directions: application to high-dimensional covariates, adaptive sampling of pairs, or hybrid losses combining MSE and rank loss.

Appendices

References